Causal Time Series Forecasting for Decision-Making

Electrolux Project

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Abstract

In collaboration with Electrolux, this project studies how Double Machine Learning (DML) can be used to estimate the causal effect of discounts on sales in retail time series data. Since discounts are assigned based on factors such as inventory, seasonality, and demand trends, standard forecasting models are affected by confounding. We therefore use a synthetic retail panel dataset with known ground-truth effects.

Across several implementations, DML correctly identifies the causal effect of discounts, corresponding to an average increase of about 1.68 units in sales volume per unit increase in discount. In contrast, naive regression and correlation-based models produce biased effect estimates and cannot be used for counterfactual analysis. Forecasting models remain responsible for prediction accuracy, while DML provides the causal information needed to evaluate pricing decisions.

We also find that the discount effect changes over time, while key demand drivers remain stable. Overall, the results show that DML complements traditional forecasting by adding a causal perspective rather than replacing predictive models.

***Index Terms*—Causal Inference, Double Machine Learning (DML), Time Series Forecasting, Retail Analytics, Confounding Bias, Pricing Strategy, Synthetic Data**

I. Introduction

Accurate sales forecasting is a core component of decision-making in retail and supply chain management, where forecasting errors can lead to inventory imbalances, inefficient pricing strategies, and lost revenue. Traditional machine learning models for demand forecasting focus primarily on predictive accuracy and typically rely on correlations in historical data. However, when forecasting is used to support pricing and promotion decisions, understanding *causal effects*, rather than correlations alone, is essential. Ignoring confounding factors may lead to biased conclusions and misleading business insights.

To address this limitation, *causal forecasting* combines causal inference with time-series prediction in order to reason about the effects of interventions such as price discounts. A central method in modern causal machine learning is *Double Machine Learning* (DML), which enables consistent estimation of causal effects in high-dimensional settings by orthogonalizing treatment and outcome models using flexible machine learning estimators [1]. By separating nuisance parameter estimation from the causal target, DML reduces bias from confounding and model misspecification. Several software frameworks provide practical implementations of DML, including the *EconML* library [2].

In this project, DML is applied to synthetic retail panel data designed to reflect realistic demand dynamics, endogenous discounting, and temporal dependence. The use of synthetic data allows controlled evaluation of causal methods under known ground-truth effects, while avoiding identification challenges present in observational retail data. In addition, standard time-series forecasting models are used as a predictive baseline to capture trend, seasonality, and autoregressive demand patterns using the *SKtime* framework [3].

By combining causal effect estimation with classical forecasting models, the project investigates how causal insights can support pricing-related decision making and how discount effects can be incorporated into future demand predictions. The remainder of this report presents the methodological framework, experimental setup, and empirical evaluation of the proposed approach.

II. Related Work

This work builds on prior research in synthetic data generation for causal modeling, Double Machine Learning for causal inference, and causal forecasting in retail applications.

A. *Synthetic Data for Causal Modeling*

Synthetic data enables controlled evaluation of causal estimation methods by providing known ground-truth treatment effects. The data generation process used in this project is based on the framework proposed by Schultz et al. [4], which describes a comprehensive simulation environment for retail demand and pricing.

Baseline demand is generated using a nonlinear function of observable covariates,

$$y_t^{(0)} = f(X_t) + \epsilon_t, \tag{1}$$

where X_t includes factors such as seasonality, stock levels, and product category. Price sensitivity is incorporated through a time-varying elasticity term,

$$y_t^{(1)} = y_t^{(0)} - \alpha_t \cdot price_t, \tag{2}$$

allowing heterogeneous responses to price changes over time and across products.

Promotional interventions are modeled using a binary treatment indicator t_t,

$$y_t^{(2)} = y_t^{(1)} + \tau_t \cdot t_t, \tag{3}$$

where the treatment effect τ_t may vary over time. To introduce realistic confounding, latent variables z_t influence both treatment assignment and demand,

$$t_t = g(z_t, X_t), \tag{4}$$

$$y_t^{(3)} = y_t^{(2)} + h(z_t). \tag{5}$$

Temporal structure is added via seasonality and trend,

$$y_t = y_t^{(3)} + A \sin\left(\frac{2\pi t}{P}\right) + Bt + \eta_t, \tag{6}$$

and the process is extended to a panel setting with multiple products i, yielding observations $(y_{i,t}, t_{i,t}, X_{i,t}, price_{i,t})$. This construction produces realistic temporal dynamics, structured confounding, and heterogeneous treatment effects, while retaining known causal parameters.

B. *Double Machine Learning for Causal Inference*

Double Machine Learning (DML), introduced by Chernozhukov et al. [1], is a framework for estimating causal effects in high-dimensional settings with strong confounding. By orthogonalizing treatment and outcome models and separating nuisance estimation from the causal target, DML reduces bias due to confounding and model misspecification.

Empirical evaluations show that DML performs well in settings with high-dimensional covariates and nonlinear relationships [5]. In this project, DML is used to estimate the causal effect of discounts on sales volume in a retail panel setting.

C. *Causal Forecasting in Retail Applications*

Recent work extends causal inference methods to time-series and panel data settings relevant for retail decision-making. Fuhr and Papies [6] show that standard DML can be biased in panel data due to unobserved unit-level heterogeneity and propose Correlated Random Effects corrections based on the Mundlak approach.

In parallel, research on causal forecasting highlights the limitations of correlation-based forecasting models under policy changes. Crasson et al. [7] demonstrate that standard forecasting models fail to generalize under interventions and propose causally motivated forecasting frameworks.

Together, these studies motivate combining causal effect estimation with time-series forecasting. The present work builds on this literature by applying DML to synthetic retail panel data to study the temporal behavior of discount effects under controlled causal conditions.

III. Problem Description

Retail decisions in pricing, inventory, and marketing are commonly supported by forecasting models. However, many key business questions are inherently *causal* rather than purely predictive, such as whether a discount increases sales, by how much, and under which conditions. Traditional forecasting methods and Marketing Mix Models (MMM) rely on historical correlations and assume that these relationships remain valid under future interventions. In the presence of confounding factors such as seasonality, stock levels, and demand trends, this assumption can lead to biased conclusions [1].

This creates a challenge for decision-making: forecasts that ignore causality may provide misleading guidance for pricing and promotion strategies. In addition, causal effects may vary across time and products, requiring decision-makers to understand not only *whether* an intervention works, but also *when* and *under which conditions* it is effective.

In collaboration with Electrolux, this project studies Double Machine Learning (DML) as a framework for causal forecasting in a retail setting. The method is evaluated using synthetically generated panel data that captures realistic retail dynamics under confounding. Causal estimation is combined with time-series forecasting to analyze the temporal stability of discount effects.

The study is guided by the following research questions:

1) How reliable are the estimated causal effects?
2) Under which scenarios does the model perform well?
3) Which parameters remain stable over time?
4) What are the implications for retail decision-making?

The overall objective is to assess whether DML can move retail forecasting beyond correlation-based prediction toward actionable, causally grounded decision support.

IV. Methodology and Implementation

This section presents the experimental pipeline, including synthetic data generation, causal effect estimation using Double Machine Learning, counterfactual forecasting, and temporal stability analysis.

A. *Synthetic Data Generation*

To evaluate causal methods under controlled conditions, we simulate a large-scale synthetic retail panel dataset with known ground-truth treatment effects. The data generation process is inspired by the framework described in Appendix E of Schultz et al. [4], but was adapted and iteratively refined during the project to ensure correct causal identification and realistic retail dynamics.

The final dataset consists of 4,500 products observed over 100 weekly time periods, resulting in 450,000 observations. Products are assigned category-specific demand parameters

and seasonal phase shifts, inducing heterogeneous demand patterns across items. An overview of the dataset configuration is provided in Table I.

TABLE I
SYNTHETIC RETAIL PANEL DATASET SPECIFICATIONS

Products	4,500
Time Horizon	100 weeks
Total Observations	450,000
Start Date	2021-01-04
Random Seed	1337
Demand Categories (α_d)	45
Base Demand Groups (β_k)	15
Seasonal Phase Groups	6
Brands / Product Lines / Groups / Subgroups	5 / 6 / 12 / 24

Demand and Price Mechanism: Observed sales volume is generated using a structural demand model,

$$q_{i,t} = q^b_{i,t} - p_{i,t} e_i, \tag{7}$$

where $q^b_{i,t}$ denotes baseline demand, $p_{i,t}$ is the transaction price after discount, and e_i is a product-specific price elasticity.

Baseline demand combines trend, seasonality, and latent capacity,

$$q^b_{i,t} = (0.15\tau_{i,t} + 0.25 s_{i,t} + 1) c_{i,t}. \tag{8}$$

The true price sensitivity e_i is drawn from a log-normal distribution and scaled by the average demand level,

$$e_i = \max(1.3, \text{LogNormal}(0.75, 0.125^2)) \cdot 0.15 \cdot \mathbb{E}[a_{i,t}]. \tag{9}$$

The main stochastic components are summarized in Table II.

TABLE II
KEY PARAMETER DISTRIBUTIONS

Trend slope γ_i	Uniform$(-0.02, 0.02)$
Elasticity e_i	LogNormal$(0.75, 0.125^2)$
Seasonal period	30 weeks
Noise $\epsilon_{i,t}$	$\mathcal{N}(0, 1)$

Endogenous Pricing and Confounding: Discounts are assigned endogenously through a stochastic, inventory-driven pricing policy, creating feedback between demand, stock levels, and future discounts. Confounding arises from several factors, including baseline demand, inventory levels, price, demand trends, and seasonality.

During the initial phase of the project, an earlier version of the data generator was implemented following the general structure above. Subsequent validation revealed structural shortcomings, including incorrect price–demand sign conventions and insufficient temporal dependence in the discount assignment mechanism. As a result, both causal estimates and forecasting behavior were inconsistent with the intended data-generating process.

These issues motivated a redesign of the discount policy and confounding structure midway through the project. The final version explicitly enforces time-dependent discount assignment and strong endogeneity, ensuring that naive correlations between discounts and sales are biased while the true causal effect remains identifiable by construction.

The resulting dataset therefore provides a realistic yet fully controlled environment for evaluating causal estimation and causal forecasting methods in retail panel data.

B. *Evaluation Criteria*

To assess both the statistical validity and practical usefulness of the proposed causal forecasting framework, three complementary evaluation criteria are used: *(i) causal accuracy*, *(ii) counterfactual consistency*, and *(iii) forecasting performance under intervention*.

Causal accuracy evaluates how well the estimated treatment effect matches the known ground-truth effect in the synthetic data by comparing the DML-estimated Average Treatment Effect (ATE) with the true structural effect.

Counterfactual consistency examines whether the model produces stable and logically consistent predictions under hypothetical policy interventions, such as removing discounts or enforcing constant discount levels, thereby testing its validity beyond historical correlations.

Forecasting performance under intervention measures predictive accuracy using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) for both baseline forecasts and DML-adjusted causal forecasts, assessing how causal adjustments affect forecasting performance under policy changes.

Together, these criteria capture predictive accuracy, causal correctness, and robustness under decision-relevant intervention scenarios.

C. *Baseline Forecasting Models*

As a reference point for evaluating the added value of causal modeling, purely predictive baseline forecasting models are constructed using the *sktime* framework. These models serve both as a benchmark for later causal analyses and as a validation that the synthetic data exhibits meaningful time-series structure rather than pure noise.

The baselines are implemented using a *recursive reduction strategy*, which reformulates the forecasting task as a supervised regression problem. Two models are considered: a *Random Forest Regressor (RF)* and a *Gradient Boosting Machine (GBM)*, chosen for their strong performance in non-linear regression settings. Input features include lagged sales and discounts, current discount level, stock, trend, seasonality, marketing activity, and seasonal sine–cosine components.

For each selected product, the data is split temporally into training and test sets using a fixed forecasting horizon of 20 weeks. Performance is evaluated using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE).

Baseline forecasting was conducted twice during the project. An initial round was performed early in the project using a preliminary version of the synthetic dataset. After identifying structural issues in the data generation process and redesigning the dataset midway through the project, the baseline models were re-estimated on the corrected data. This iterative procedure ensured that subsequent causal analyses were evaluated

against a consistent and structurally valid predictive benchmark.

Overall, the baseline models provide a purely predictive point of comparison against which the causal forecasting approach based on DML is assessed.

D. DML Estimation Pipeline

The causal effect of discounts on sales volume is estimated using multiple implementations of Double Machine Learning (DML) to ensure robustness and methodological validity. Across all approaches, the treatment is defined as the continuous discount level, the outcome is sales volume, and the confounder set includes trend, seasonality, marketing activity, stock levels, lagged demand, and pricing variables. All models estimate the Average Treatment Effect (ATE) of a marginal change in discount.

The initial objective of the project was to evaluate a time-series variant of DML implemented in the *sktime* framework and assess whether it could recover the known treatment effect embedded in the synthetic data. To establish a causal reference, a custom vanilla DML estimator was first implemented. While vanilla DML successfully recovered the ground-truth effect, the *sktime*-based DML produced unstable and biased estimates. This discrepancy motivated further validation using alternative DML implementations to confirm the correctness of the custom estimator and to better understand the limitations of time-series DML in this setting.

a) ***Vanilla DML****:* A standard cross-fitted DML estimator is implemented using five-fold sample splitting. Random Forest regressors are used for both nuisance components, the treatment model $m(X) = \mathbb{E}[D|X]$ and the outcome model $g(X) = \mathbb{E}[Y|X]$. Orthogonalized residuals are constructed and the ATE is estimated via residual-on-residual regression. This estimator serves as the primary causal baseline.

b) ***EconML Validation****:* To validate the custom implementation, the analysis is replicated using *LinearDML* from the *EconML* library. In addition, *CausalForestDML* is applied to test robustness under nonlinear and heterogeneous treatment effects. These models provide an external reference and support validation of the vanilla DML results.

c) ***Panel DML with Mundlak Correction****:* Given the panel structure of the data, unobserved time-invariant product characteristics may bias standard DML estimates. To address this, a Mundlak (Correlated Random Effects) transformation is applied by augmenting the confounder matrix with product-level means of time-varying covariates. This CRE-DML specification improves stability and accounts for unobserved heterogeneity.

d) ***Time-Series DML using sktime****:* A time-series adaptation of DML is implemented using *sktime* within a recursive forecasting setup. In this setting, the estimated effects are highly unstable and deviate substantially from the known ground truth. This behavior is attributed to strong temporal dependence and short rolling windows, which complicate causal identification. The *sktime*-DML results are therefore used for methodological comparison rather than final inference.

Overall, the causal analysis follows an iterative validation strategy: vanilla DML establishes a baseline estimate, EconML implementations provide external validation, Causal Forest assesses robustness to nonlinearity, and Mundlak-DML addresses panel-specific bias. This process ensures that the final causal estimates are both statistically and structurally sound.

E. Item Selection, Counterfactual Forecasting, and Stability Analysis

To evaluate the causal forecasting framework across heterogeneous settings, a representative subset of products was selected from the full panel dataset based on three criteria: sales volume (low, medium, high), average discount level (low, medium, high), and seasonal sensitivity (low vs. high correlation with seasonality). The final evaluation set is defined as the union of all selected items, ensuring diversity in demand patterns and pricing regimes.

For each item, forecasting is evaluated over a fixed horizon of 20 weeks. Since the data is synthetic, true future outcomes are known, allowing direct comparison between predictive and causal forecasts. Baseline forecasts are generated under the historical policy using the *sktime* model. Causal forecasts are then constructed by adjusting the baseline predictions using the estimated DML effect,

$$\hat{y}_t^{\text{causal}} = \hat{y}_t^{\text{base}} + \hat{\theta} \cdot T_t,$$

where $\hat{\theta}$ denotes the estimated causal effect of discounts.

To assess counterfactual behavior, two policy scenarios are considered: a no-discount scenario ($T_t = 0$) and a constant-discount scenario ($T_t = 0.5$). Baseline and DML-based forecasts are compared under both historical and counterfactual policies to evaluate demand responses to pricing changes.

Finally, temporal stability is analyzed by re-estimating DML effects over early, mid, and late time periods. Feature stability is further assessed using Ordinary Least Squares coefficients and Random Forest feature importances, allowing identification of stable demand drivers and time-varying effects.

V. Experimental Evaluation

This section presents the empirical results, including validation of the synthetic data, estimation of causal discount effects using DML, and comparison across model variants.

A. Synthetic Data Overview

We first validate the properties of the synthetic dataset. Figure 1 shows the correlation structure between key variables. Discounts are positively correlated with sales volume, while strong confounding relationships are present, including a negative correlation between price and discounts and a positive correlation between seasonality, discounts, and volume. This confirms that naive regression estimates are unreliable.

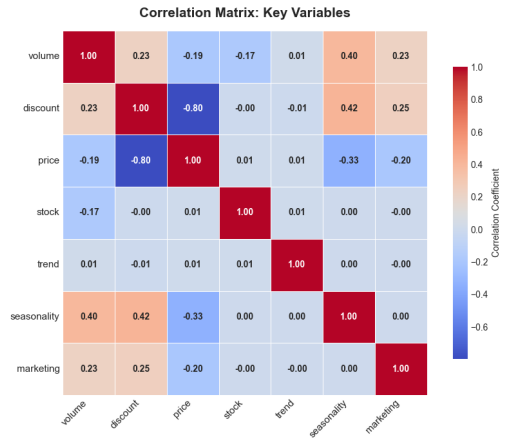

Fig. 1. Correlation matrix between key variables in the synthetic dataset.

Figure 2 illustrates the ground-truth causal structure used in the data-generating process. Discounts affect sales both directly and indirectly through price, while stock, seasonality, trend, and marketing act as confounders. Heterogeneous treatment effects introduce realistic variation across products.

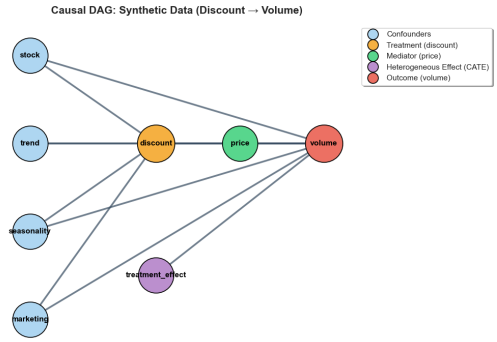

Fig. 2. Causal Directed Acyclic Graph (DAG) of the data-generating process.

Figure 3 compares observed sales volume with the structural prediction implied by the true causal model. The close alignment confirms that the ground-truth treatment effect is consistently embedded in the synthetic data.

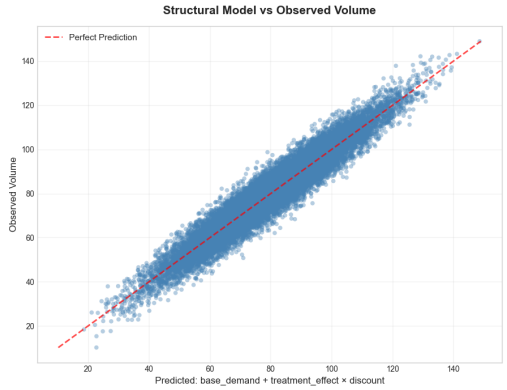

Fig. 3. Observed volume versus structural prediction from the true data-generating process.

B. *Causal Effect Estimation using DML*

Figure 4 shows the distribution of true item-level treatment effects, with an average ground-truth effect of

$$\mathbb{E}[\tau_i] = 1.684,$$

indicating substantial heterogeneity across products.

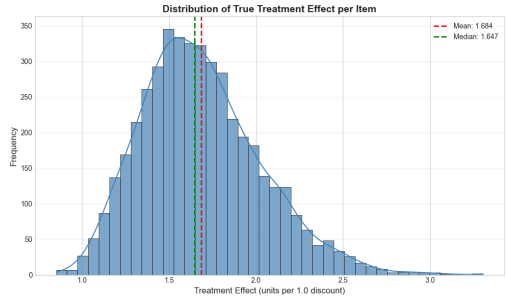

Fig. 4. Distribution of the true treatment effect per item.

TABLE III
ESTIMATED CAUSAL EFFECT OF DISCOUNT ON SALES

Method	ATE	95% CI
True effect (synthetic)	1.684	–
Vanilla DML	1.686	[1.458, 1.914]
EconML LinearDML (average)	1.646	[0.995, 2.297]
EconML CausalForestDML	1.589	[−13.46, 16.64]
CRE / Mundlak Panel-DML	1.889	[1.720, 2.058]
sktime Time-Series DML	16.699	–

Table III compares estimated treatment effects across DML variants. Vanilla DML, LinearDML, and CRE Panel-DML recover the true effect accurately. CausalForestDML exhibits high variance, while the time-series DML implementation substantially overestimates the effect. These results confirm that standard and panel-adjusted DML provide reliable causal estimates in the synthetic retail setting.

C. *Forecasting under Intervention and Counterfactual Scenarios*

This subsection evaluates how the estimated causal discount effect influences forecasting under historical and counterfactual scenarios, based on average behavior across 40 representative items.

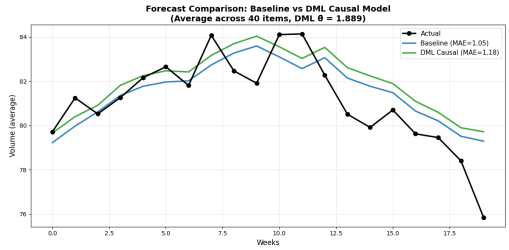

Fig. 5. Average forecast comparison across 40 items. Baseline sktime forecasts, DML-adjusted forecasts, and true sales over the 20-week test horizon.

Figure 5 shows that both baseline and DML-adjusted forecasts track the overall sales trend. The baseline model achieves slightly lower average error (MAE = 1.05) than the causal model (MAE = 1.18), indicating that incorporating causal information does not improve short-term accuracy under the historical policy.

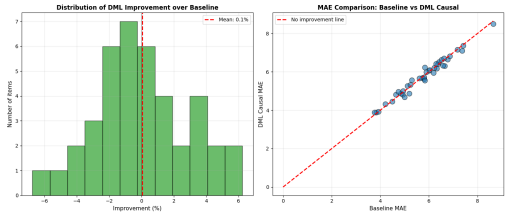

Fig. 6. Left: Distribution of relative MAE improvement of DML over the baseline. Right: MAE comparison per item.

At the item level (Figure 6), the mean relative improvement is close to zero (0.06%), with 20 out of 40 items showing lower error under the causal model. This indicates heterogeneous effects and no consistent gain in predictive accuracy.

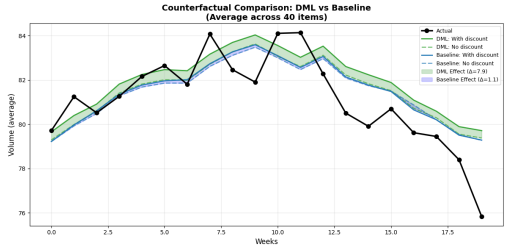

Fig. 7. Counterfactual comparison between baseline and DML forecasts under discount and no-discount scenarios.

The key difference emerges under counterfactual interventions (Figure 7). When discounts are removed, the baseline model implies a small average effect (1.1 units), while the DML-based model predicts a substantially larger response (7.9 units), highlighting that baseline forecasts capture correlations whereas DML isolates causal effects.

Overall, DML primarily improves *counterfactual validity* rather than short-horizon forecast accuracy, providing more realistic estimates of promotional impact.

D. *Temporal Stability of Causal Effects and Model Parameters*

To assess temporal robustness, the panel is divided into early, mid, and late periods. DML is re-estimated for each period, and stability is analyzed using standardized OLS coefficients and Random Forest (RF) feature importances.

a) Discount Effect Stability.: Period-specific ATE estimates vary over time:

$$\text{ATE}_{early} = 1.41, \quad \text{ATE}_{mid} = 1.84, \quad \text{ATE}_{late} = 0.97,$$

with all confidence intervals excluding zero. This indicates that discount elasticity is statistically significant but time-varying.

b) Stability of Demand Drivers.: OLS coefficients show that lagged volume, marketing, and seasonality remain stable across periods, while stock and weeks_left exhibit notable drift (Figure 8). RF importances confirm high stability in predictor ranking, with lagged volume consistently dominating (Figure 9).

c) Summary.: While the causal effect of discounts varies over time, core demand drivers remain stable. This suggests that baseline demand can be forecast reliably, but discount elasticity should be treated as dynamic, motivating periodic re-estimation of causal effects.

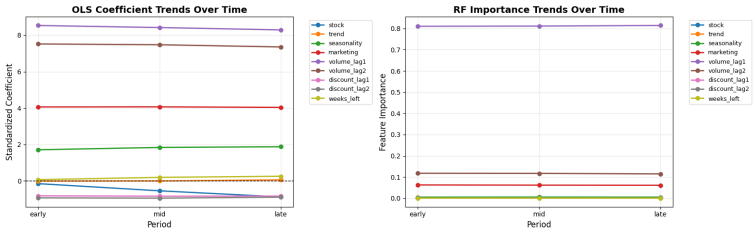

Fig. 8. Temporal evolution of OLS coefficients (left) and RF feature importances (right).

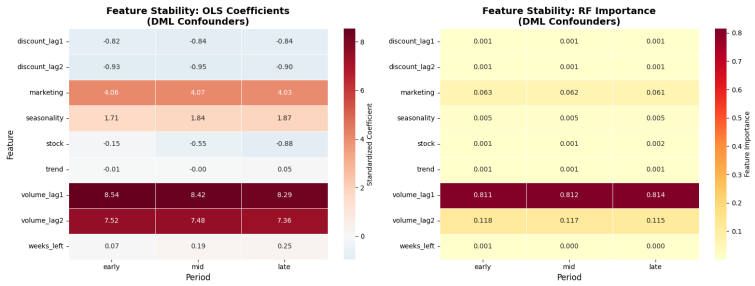

Fig. 9. Stability of standardized coefficients across periods.

VI. Summary and Discussion

A. *Summary*

This study applied Double Machine Learning (DML) to synthetic retail panel data to estimate the causal effect of discounts and support intervention-valid analysis. Across multiple implementations, including vanilla DML, EconML's LinearDML, and a Mundlak-style CRE specification, DML consistently recovered the true causal effect (ATE $\approx$ 1.68). In contrast, baseline forecasting models achieved slightly better short-term predictive accuracy but failed to generalize under counterfactual interventions.

Temporal stability analysis showed that core demand drivers such as lagged sales, marketing, and seasonality remain stable over time, while the discount effect varies across periods, indicating regime-dependent promotional elasticity. Overall, the results show that DML complements predictive models by providing the causal structure needed to evaluate pricing strategies and support prescriptive decision making.

B. Discussion

The results show that Double Machine Learning (DML) provides reliable causal effect estimates in a retail setting with strong confounding and heterogeneous demand dynamics. Vanilla DML, LinearDML, and the CRE-based panel specification consistently recover the true structural effect, indicating that the orthogonalization framework is well suited for retail panel data where treatment assignment depends on both observed and latent factors.

A substantial portion of the project time was devoted to the design and validation of the synthetic data generator. Early versions of the dataset contained structural shortcomings, including insufficient temporal dependence and incorrect discount dynamics, which led to misleading causal and forecasting results. Correcting these issues required redesigning the data-generating process midway through the project and re-running large parts of the analysis. This iterative process delayed overall progress but was necessary to ensure valid causal identification.

In contrast to standard DML implementations, the time-series DML variant implemented using *sktime* produced unstable and biased estimates. This behavior is attributed to strong cross-item heterogeneity and item-specific dynamics in the synthetic panel, which violate the homogeneous time-series assumptions underlying recursive forecasting approaches. As a result, causal residuals were inconsistent across items, leading to unreliable estimates.

Importantly, this study does not demonstrate that DML improves forecasting accuracy in this setting. While predictive models achieved slightly better short-term accuracy under historical policies, DML successfully recovered the true causal effect of discounts. The primary value of DML in this study is therefore causal identification rather than improved prediction. In addition, DML was a new methodology for the authors, and significant effort was required to understand its assumptions, implementation, and how causal effects should be incorporated into forecasting. This learning process likely influenced both modeling choices and empirical results.

Overall, the findings highlight the complementary roles of predictive and causal models. Forecasting models remain effective for short-term prediction under stable policies, while DML provides the causal insights required to evaluate pricing interventions and counterfactual scenarios. Integrating DML as a causal layer on top of traditional forecasting systems offers a more robust foundation for data-driven retail decision making.

VII. Summary of discussions with the opposing group

The opposing group applied Double Machine Learning (DML) to a real-world retail dataset to estimate the causal effect of discounts on demand. Their work focused on identifying confounding patterns, evaluating the temporal stability of discount effects, and comparing naive correlation-based models with causally adjusted estimates. They found that DML consistently revealed positive promotional effects that standard predictive models failed to capture. Overall, their analysis emphasized the practical relevance of causal inference for real retail pricing decisions.

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